be no common path till SEP realised that the M-40 engine cannot be upgraded beyond a specific limit for LIIIS, the proposed geosynchronous satellite launch vehicle. This meant developing a new engine, and here came our opportunity to strike a deal.

But it wasn't easy as SEP wanted money, and ISRO didn't have much to offer. Kanwal Grover, who became an SEP agent, worked on the deal more for his love for India than anything else. Such was Grover's commitment to the national interest that he forwent his commission to make the deal work. When he saw a possible negotiation over joint development of the new engine, Grover asked me bluntly how much money India could spend on the deal.

'Not a paisa,' I said.

'Then how? The French want money.'

'We will do a barter.'

I didn't realise what I had just said. A barter? What do we have to give in return? We are still a bunch of scientists firing sounding rockets and struggling with a three-ton liquid rocket engine, which appeared to be a toy against the 60-tonner the French were about to develop. But that triggered a chain of thoughts in me, eventually leading to a contract so ingenious that we partnered to develop the Viking-Vikas engine along with the French with no money going out of India. What's more, India got a transducer-making facility, which is still churning out the tiny pressure sensors used in rockets, with machines and know-how from France.

Here is how we made it.

I told Grover that India will not be able to spend even a rupee in cash, but our manpower could be bartered for the technology. About fifty of our engineers would come and work in France along with SEP men to develop the engine. Grover laughed.

'As if you have so many engineers good enough for the job,' he said.

Again, without thinking I said, 'We have much more than fifty fine engineers.'

I was trying to convince myself than Grover, who soon agreed to take up the proposal with Souffle and Morin. The SEP heads found merit in the